

Sweep:

$\Phi_M - \Phi_{PZC}$ (V):

Φ_{min} (V):

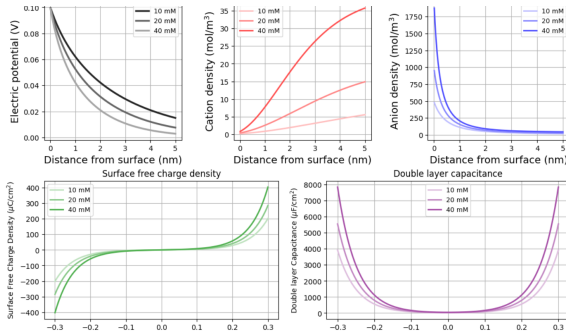
Φ_{max} (V):

x_{max} (nm):

Base values used for parameters not being swept:

base T:

base ϵ_r :



EXERCISE 2

Electric Double Layer Modeling: Gouy–Chapman

April 19, 2026 | O. Schmidt | Theory and Computation of Energy Materials (IET-3)

AIM OF TODAY'S EXERCISES

Today, we investigate the Gouy–Chapman model. Some help can be found on

→ [Exercises](#) ←

Objectives: To understand the strengths and limitations of the Gouy–Chapman model with respect to

- Electric potential
- Anion and Cation concentration profile
- Surface Free Charge Density
- Double Layer Capacitance

GOUY-CHAPMAN: PARAMETRIC SETUP

Fundamental constants			Model parameters		
Symbol	Value	Unit	Symbol	Unit	Meaning
N_A	6.022×10^{23}	mol^{-1}	$\phi_M - \phi_{\text{pzc}}$	V	Applied potential at metal surface relative to the Potential of Zero Charge (PZC).
k_B	8.617×10^{-5}	eV K^{-1}	T	K	Temperature
e_0	1.602×10^{-19}	C	ε_r	–	Relative permittivity
ε_0	8.854×10^{-12}	F m^{-1}	c_b	mM	Bulk ion concentration

MODEL SETUP: GOUY–CHAPMAN

The differential equation system describing the Gouy–Chapman model:

Gouy–Chapman: Governing equation for ϕ

$$\text{GC (1D): } \begin{cases} \frac{d^2\phi}{dx^2} = -\frac{2e_0 N_A c_b}{\varepsilon_r \varepsilon_0} \sinh\left(\frac{e_0 \phi}{k_B T}\right), & x \in]0, \infty[\\ \phi(x=0) = \phi_M - \phi_{\text{pzc}}, & x = 0 \\ \lim_{x \rightarrow \infty} \phi(x) = 0, & x \rightarrow \infty \end{cases}$$

with

$$c_-(x) = c_b \exp\left(\frac{e_0 \phi(x)}{k_B T}\right), \quad c_+(x) = c_b \exp\left(-\frac{e_0 \phi(x)}{k_B T}\right)$$

being the concentration profiles of the anion (c_-) and cation (c_+).

MODEL SETUP: SOLUTION

Gouy–Chapman: Solution

$$\phi^{(\text{GC})}(x) = 4k_{\text{B}} T e_0^{-1} \operatorname{arctanh} \left(\tanh \left(\frac{e_0 (\phi_{\text{M}} - \phi_{\text{pzc}})}{4k_{\text{B}} T} \right) \exp(-\lambda_{\text{D}}^{-1} x) \right)$$

with $\lambda_{\text{D}}^{-1} = \left(\frac{2e_0^2 N_{\text{A}} c_{\text{b}}}{\varepsilon_0 \varepsilon_r k_{\text{B}} T} \right)^{\frac{1}{2}}$ being the inverse Debye Length.

MODEL SETUP: QUESTIONS

- 1.1) How does the electric potential change with varying $(\phi_M - \phi_{pzc})$?
- 1.2) At which $(\phi_M - \phi_{pzc})$ does the ϕ flip?
- 1.3) How do the concentrations change with varying $(\phi_M - \phi_{pzc})$, respectively?
- 1.4) At which $(\phi_M - \phi_{pzc})$ does the concentrations flip?
- 1.5) Comment on the concentration densities near the metal surface.

Input: Applied Potential ▾

Sweep: -0.1, 0.0, 0.1

Φ_{min} (V): -0.3

Φ_{max} (V): 0.3

x max (nm): 5

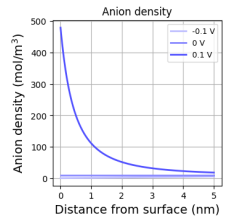
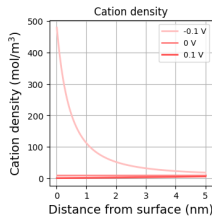
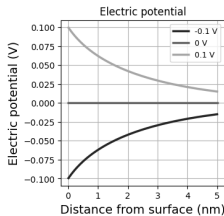
Base values used for parameters not being swept:

base c: 10

base T: 298.15

base ϵ_r : 78.5

Plot



SURFACE CHARGE: GOUY–CHAPMAN

Surface Free Charge density σ_{free}

$$\sigma_{\text{free}} = -\epsilon_0 \epsilon_r \left. \frac{d\phi}{dx} \right|_{x=0^+}$$

$$\sigma_{\text{free}} (\phi_M - \phi_{\text{pzc}}) = 0 \iff \phi_M = \phi_{\text{pzc}}$$

Surface Free Charge density σ_{free} : Gouy–Chapman

$$\sigma_{\text{free}}^{(\text{GC})} (\phi_M - \phi_{\text{pzc}}) = \sqrt{8k_B T \epsilon_0 \epsilon_r c_b} \sinh \left(\frac{e_0 (\phi_M - \phi_{\text{pzc}})}{2k_B T} \right)$$

CAPACITANCE: GOUY-CHAPMAN

Double Layer Capacitance C_{dl}

$$C_{dl}(\phi_M - \phi_{pzc}) = \frac{d\sigma_{free}}{d(\phi_M - \phi_{pzc})}$$

Double Layer Capacitance C_{dl}

$$C_{dl}^{(GC)}(\phi_M - \phi_{pzc}) = \varepsilon_0 \varepsilon_r \lambda_D^{-1} \cosh\left(\frac{e_0(\phi_M - \phi_{pzc})}{2k_B T}\right)$$

σ_{free} AND C_{dl} : GOUY-CHAPMAN

- 2.1) Show that if $\phi_M = \phi_{\text{pzc}}$, then $\sigma_{\text{free}}^{(\text{GC})} = 0$. Hint: $\sinh(0) = 0$.
- 2.2) Derive $C_{\text{dl}}^{(\text{GC})}(0)$. Hint: $\cosh(0) = 1$
- 2.3) Using the provided code and/ or the figure below, assess to which extend the GC model agrees with the experimental data.

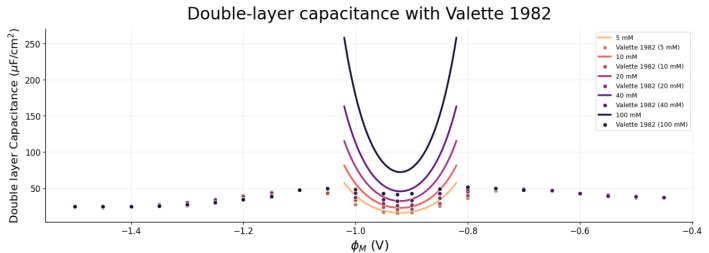
T (K):

ϵ_r :

ϕ_{pzc} (V):

ϕ_{min} (V):

ϕ_{max} (V):



Valette, G. (1982). Double layer on silver single crystal electrodes in contact with electrolytes having anions which are slightly specifically adsorbed: Part ii. the (100) face. *Journal of Electroanalytical Chemistry and Interfacial Electrochemistry*, 138(1), 37–54

OVERALL MODEL ASSESSMENT: QUESTIONS

3.1 What are the shortcomings of the Gouy–Chapman model?

3.2 What are the strengths of the Gouy–Chapman model?

capillary maximum, is observed to be -0.24 volt (table 1). If one assumes that this is also the value of ψ^o , then equation 72 gives for η_+^d a value of $192 \mu \text{ coul./cm.}^2$ (an impossible result). This is to be compared with the experimental value of $9.1 \mu \text{ coul./cm.}^2$. At higher concentrations the disagreement is even worse and the calculated value even more impossible ($4300 \mu \text{ coul./cm.}^2$ at $1 M$). These results illustrate the necessity of distinguishing between ψ^i and ψ^o , and also

Grahame, D. C. (1947). The electrical double layer and the theory of electrocapillarity. *Chemical reviews*, 41(3), 441–501